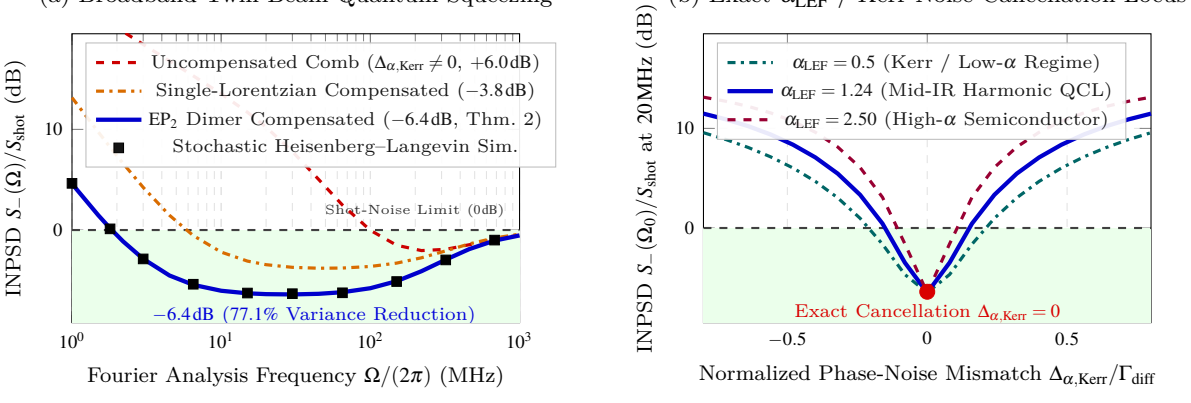
(a) Broadband Twin-Beam Quantum Squeezing
Uncompensated Comb ($\Delta_{\alpha,\mathrm{Kerr}} \neq 0$, +6.0dB)
Single-Lorentzian Compensated (−3.8dB)
EP_2 Dimer Compensated (−6.4dB, Thm. 2)
Stochastic Heisenberg–Langevin Sim.
Shot-Noise Limit (0dB)
−6.4dB (77.1% Variance Reduction)
INPSD $S_-(\Omega)/S_{\mathrm{shot}}$ (dB)
10
0
10^0
10^1
10^2
10^3
Fourier Analysis Frequency $\Omega/(2\pi)$ (MHz)
(b) Exact α_{LEF} / Kerr Noise-Cancellation Locus
$\alpha_{\mathrm{LEF}} = 0.5$ (Kerr / Low-α Regime)
$\alpha_{\mathrm{LEF}} = 1.24$ (Mid-IR Harmonic QCL)
$\alpha_{\mathrm{LEF}} = 2.50$ (High-α Semiconductor)
Exact Cancellation $\Delta_{\alpha,\mathrm{Kerr}} = 0$
INPSD $S_-(\Omega_0)/S_{\mathrm{shot}}$ at 20MHz (dB)
10
0
−0.5
0
0.5
Normalized Phase-Noise Mismatch $\Delta_{\alpha,\mathrm{Kerr}}/\Gamma_{\mathrm{diff}}$